



Understanding consumer perceived value of casual sportswear: An empirical study

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ARTICLE INFO

Available online 13 July 2011

Keywords:

Casual sportswear
Consumer perceived value
Empirical study

ABSTRACT

Value creation is the key for business to achieve a long-term success. Using a consumer perceived value (CPV) model consisting of four constructs—*price value*, *quality value*, *social value*, and *emotional value*, this study empirically assessed the CPV of casual sportswear in the U.S. market. Based on the primary data gathered by a nationwide survey, through exploratory and confirmatory factor analyses, the model was proven valid and the four constructs accounted for the most variance of CPV. Compared to social and emotional values, price showed the most influential impact on the overall CPV, while quality was also perceived as very important.

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1. Introduction

The U.S. is the world's largest sportswear market, accounting for some 40% of global sales by value. With estimated sales of \$61 billion in 2005, the sportswear market makes up about 28% of overall U.S. apparel consumption (STS market research, 2008). According to the Sporting Goods Manufacturers Association (SGMA) (2008), the sportswear market comprises of three segments: namely, active sports clothing (apparel designed and specifically purchased for use in active sports), sports clothing (fitness-oriented apparel bought for general lifestyle usage), and licensed sports clothing (lifestyle apparel bearing team or league logos). The latter two segments are usually referred to casual sportswear compared to the first segment and target the mass consumer market. In general, casual sportswear includes everything from basic items, such as inexpensive sports socks, to high-end designer pieces, such as golf shirts.

As one of the largest cotton producers in the world, the growth of the U.S. cotton industry hinges on the development of the apparel market, since over 60% of cotton fibers are used in apparel products. According to Cotton Incorporated (2009), cotton usage in casual sportswear continues to grow faster than in other apparel categories, at an average 5% per annum in the last few years. This phenomenon is primarily due to increased consumption in some major categories such as t-shirt, sweaters, and fleece jackets. In 2008, more than five billion t-shirts and nearly one billion sweaters were sold solely in the U.S. market, which amount on average to 16 t-shirts and three

sweaters purchased for each American (AAFA Annual Statistic, 2009). Based on the sales data and growth rate, it is evident that the recent consumption trends of casual sportswear provide lucrative opportunities for U.S. cotton producers and marketers. However, despite the size and importance of the market and the interest shown by industrial practitioners, timely and systematic scholarly research devoted to understanding this scenario has been lagging. Although some relevant analyses have been conducted by various market research companies (e.g., Cotton Incorporated, Mintel), most of these works are descriptive in nature and are highly dependent on subjective judgment. There is a lack of well-designed academic studies, which are theory-led and use rigorous analytical models, and sophisticated statistical methods.

In order to address the gap in the literature and provide essential understanding of the market, this paper aimed to empirically assess the consumer perceived value (CPV) of casual sportswear in the U.S. market. Value is considered to be a key constituent of marketing and the ability of a company to provide desired value to its customers has become a means of differentiation and the reason for a company's existence as well as for its success (Ravald and Gronroos, 1996; Slater, 1997). The analytical model and corresponding measures and scales used were adopted from the widely applied Sweeney and Soutar (2001) CPV framework and were slightly refined to reflect this product specific study. Consumers' shopping habits and knowledge and their perception of cotton versus other common fibers in the context of casual sportswear were also investigated. The primary data were gathered via a nation-wide consumer questionnaire survey. Exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) within structural equation modeling (SEM) were applied to conduct the statistical analysis. The CPV driving U.S. consumers' attitudes and possible purchase intentions toward casual sportswear were quantitatively determined.

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2. Literature review

Prior studies have shown that there are discrepancies between what companies think their consumers value and what consumers really value (e.g., [Parasuraman et al., 1985](#)). Therefore, understanding of CPV has become a strategic imperative for companies and a major focus of interest in the marketing literature ([Parasuraman and Grewal, 2000](#); [Sánchez-Fernández and Iniesta-Bonillo, 2007](#)). The number of CPV related research studies has been growing rapidly in the last two decades. Empirical findings have revealed that CPV is even more important than consumer satisfaction in explaining consumer purchase behavioral intentions, as CPV occurs through various stages of the purchase decision-making process, including the pre-purchase stage while consumer satisfaction is unanimously agreed to be a post-purchase or post-use evaluation ([Tam, 2004](#)). [Woodruff \(1997\)](#) stressed that being able to have products better meet CPV compared with competitors is a major source of competitive advantage.

Although the theoretical foundation of CPV is still under development, there are two theories, namely means-end theory and the theory of market choice behavior, which have been extensively discussed in relation to CPV. Means-end theory was initially proposed to describe how consumers cognitively categorize information about products ([Gutman, 1982](#)). This theory argues that CPV is formed by assessing how well a product's attributes perform relative to consumer desires ([Petrick and Backman, 2002](#)). Compared to means-end theory, the theory of market choice behavior, which was first introduced by [Sheth et al. \(1991\)](#), is focused on consumption value (i.e., functional, social, emotional, epistemic, and conditional value) relative to various consumption circumstances and product types. This theory holds that market choice is a function of multi-dimensional consumption values, these values affect consumer consumption behavior differently in various circumstances, and these values are independent of one another. This theory suggests that perceived and real value for consumption can predict consumer attitudes and behavior toward purchase. The theory of market choice behavior provides a strong theoretical foundation from which to build CPV measures and scales ([Sweeney and Soutar, 2001](#)).

The definitions of value in the marketing literature have been evolving over time. [Zeithaml \(1988\)](#) stated that perceived value is consumers' overall assessment of the utility of a product or service based on their perceptions of what is received versus what is given up. The early definitions are mainly focused on the trade-off decision between two primary dimensions, namely quality and price ([Monroe, 1990](#)). For different consumers, quality and price are differentially weighed. Some consumers perceive value when they pay low price while others perceive value when there is a balance between quality and price.

Starting from the 1990s, multi-dimensional conceptualization and operationalization of CPV have been gaining popularity. More dimensions have been discussed and added to enhance the understanding of CPV, since a trade-off between only quality and price was generally considered too simplistic and incomplete ([Woodruff, 1997](#)). In the meantime, the measures and scales for quantifying CPV in distinct dimensions have been proposed and tested by an increasing number of studies.

[Sheth et al. \(1991\)](#) contended that consumer choice is a function of multi-dimensional consumption value and these dimensions make varying contributions in different choice situations. Five dimensions, namely social, emotional, functional, epistemic, and conditional value, were proposed to measure CPV, at either the decision to buy level, the product comparison level, or the brand comparison level. Work by [Sheth et al. \(1991\)](#) provided a springboard to many further measure development studies.

[Kantamneni and Coulson \(1996\)](#) proposed a four-dimensional model to measure CPV, namely societal value, experiential value, functional value, and market value. [Grewal et al. \(1998\)](#) used different terms to group CPV into two dimensions: perceived acquisition value which focuses on good value for the money, and perceived transaction value, which is about the pleasure of finding a good deal. They developed corresponding measures and scales for each of these dimensions. Taking a similar approach as [Grewal et al. \(1998\)](#) and [Parasuraman and Grewal \(2000\)](#) added in-use value and redemption value into the CPV model, in addition to acquisition value and transaction value. The post-purchase value evaluation was also incorporated as part of the CPV model in their paper, although CPV was often assessed at a pre-purchase stage and was used to generate useful information for product development and marketing strategies.

As consumer research evolved from a focus on the cognitive aspects of the decision-making process to include more intrinsic aspects of a product, [Sweeney and Soutar \(2001\)](#) indicated that there is a need for an enhanced CPV model with better designed measures which can more accurately and completely identify the important values perceived by consumers at a product level or a service level. Through factor analysis using the consumer survey data gathered at both pre- and post-purchase stages, they identified four distinct dimensions of CPV and specially termed them as PERVAL. They are emotional value, social value, quality/performance value, and price/value for money, all of which were found to have a significant explanatory power for consumer attitudes and purchase behavior. Since [Sweeney and Soutar's \(2001\)](#) CPV study was published, it has been widely cited or applied in numerous conceptual and empirical studies. Some researchers have suggested that minor revisions or enhancements could be done to make PERVAL measures more effective in capturing CPV, given a specific product or service. However, due to its theoretical parsimony, high accountability of CPV, and superior applicability in empirical analysis (e.g., [Yang and Peterson, 2004](#)), [Sweeney and Soutar's \(2001\)](#) PERVAL measures have been generally accepted as one of the best tools to empirically investigate perceived values from a consumer perspective.

[Yang and Peterson \(2004\)](#) applied PERVAL along with consumer satisfaction measures to analyze the effect of CPV in a setting of internet retailing. [Petrick \(2004\)](#) incorporated CPV into an empirical study of the relationships between consumer satisfaction and perceived value, and their predictability of repurchase intentions and positive word of mouth publicity. [Tam \(2004\)](#) applied the concept of CPV to the restaurant service industry in Hong Kong, and proposed an integrative model to examine the relationships among CPV, consumer satisfaction, and post-purchase behavior. [Pura \(2005\)](#) analyzed the direct effect of CPV dimensions on consumers' attitudinal and behavioral components of loyalty for location-based mobile services. In a similar fashion, [Cheng et al. \(2009\)](#) investigated the impact of perceived value on customer intention to use the internet as a retailing platform. Given the fact that CPV leads to higher purchase and word-of-mouth, [Dahlén et al. \(2009\)](#) studied the marketing means to enhance CPV of brands.

The above literature review shows that the knowledge about CPV has been growing rapidly, especially in the last decade, and the empirical application of CPV to specific products or services has been regarded as a key to the question of how a company can determine what consumers want and achieve a sustainable competitive advantage ([Ravald and Gronroos, 1996](#)). Against this backdrop, many researchers have argued that accurate measurement of CPV is one of most significant advances in the marketing literature in recent times, deserving considerably more attention than it has been thus far given ([Sánchez-Fernández and Iniesta-Bonillo, 2007](#)). This situation clearly supports the need for research applying CPV to specific products of interest to better understand the values perceived by consumers.

3. Methodology

3.1. Analytical model and survey instrument development

CPV for casual sportswear was conceptualized as a multi-dimensional construct. An analytical model was developed based on Sweeney and Soutar (2001)'s framework. Four dimensions in the model, namely emotional value, social value, quality value (performance value), and price value (value for money), aimed to measure the value of casual sportswear perceived by U.S. consumers. Fig. 1 illustrates the proposed model in an SEM format for

statistical analysis purposes. The dimensions of CPV were operationalized into four first-order latent constructs to determine the second-order latent construct CPV of casual sportswear. When there are several constructs, which can be meaningfully conceptualized at a higher order of abstraction, a higher-order modeling approach would be the most suitable technique that can represent such structures (Koufteros et al., 2009).

In addition, questions to elicit information about general attitudes towards cotton in casual sportswear were included in the questionnaire. Their purpose was to obtain information that would aid U.S. cotton producers, and casual sportswear

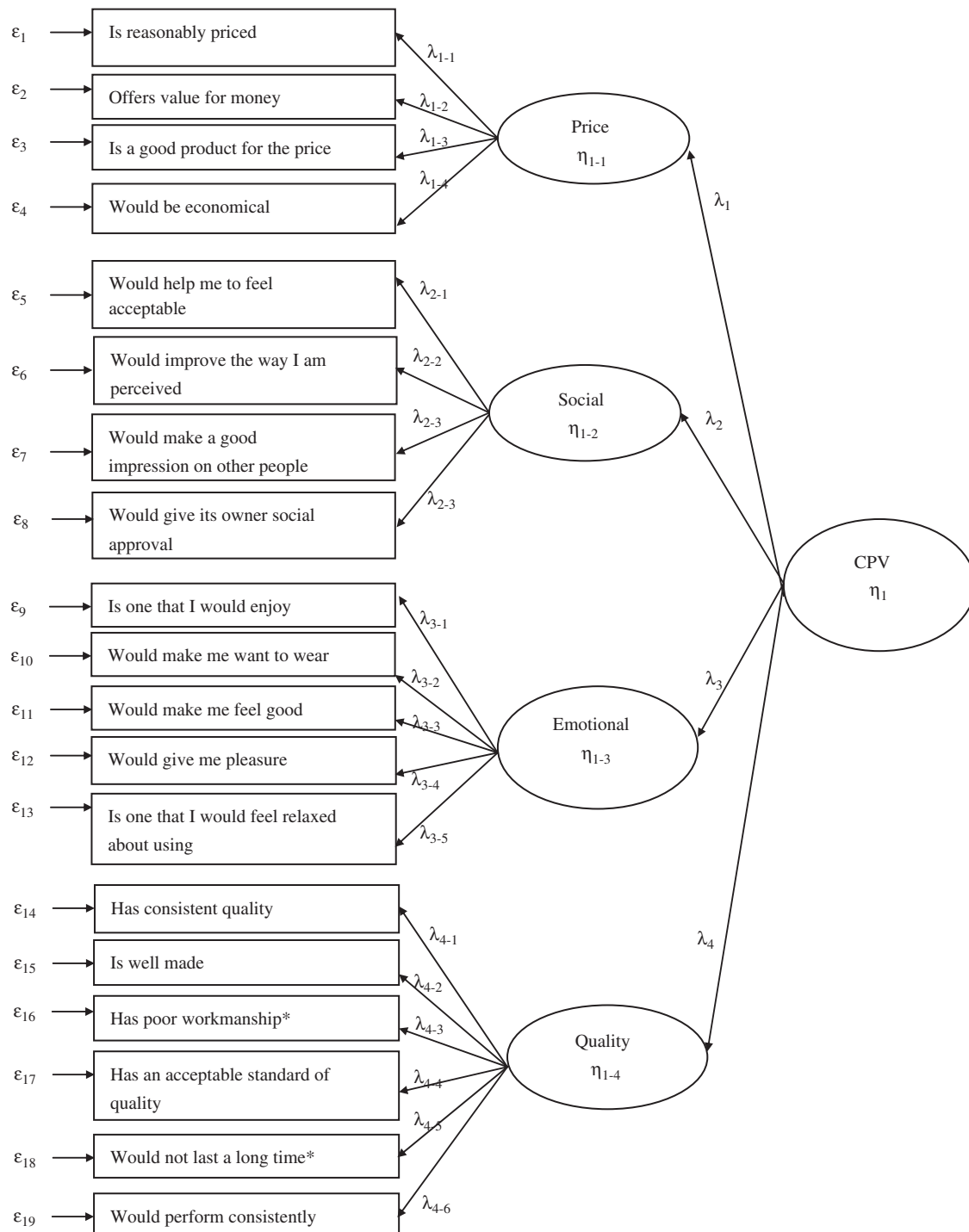


Fig. 1. The proposed CPV model for casual sportswear. *Note:* the model is developed based on Sweeney and Soutar (2001)'s framework. A seven-point Likert scale (from 1=extremely unimportant to 7=extremely important) for the measurement of variables in each latent construct. * reverse scored.

manufacturers and retailers in the improvement of product development and marketing of cotton in casual sportswear.

3.2. Survey

The developed survey instrument was first reviewed by experienced professors in the field and then pre-tested using college student responses. The instrument was thus refined with regard to content, arrangement, wording accuracy, and relevance. This procedure helped make the final survey instrument more valid and clearer.

At the completion of the pre-test stage, online questionnaire survey format was chosen for the main data collection, since this allowed relatively inexpensive access to a large number of consumers who were spread across the U.S. The survey was conducted in cooperation with a professional market research company in order to get a higher number of returns. The web-based questionnaire was hosted on a professional online survey server. The invitations for online survey participation were sent out by the market research company to its national consumer database, which resulted in a total of 5195 returns. Among the 5195 returns, 246 returns were dropped due to the significant number of incomplete questions. Finally, there were 4949 returns eligible for further statistical analysis. Because the number of consumers approached during the survey invitation stage was unknown, the response rate was not available. However, the number of returns was ideal for sophisticated statistical analysis.

3.3. Statistical methods

Non-response bias was evaluated using the *t*-test on demographic variables. As a convention, the responses of early and late groups of returned surveys were compared to provide support of non-response bias (Lambert and Harrington, 1990). Exploratory factor analysis (EFA) is recommended to identify measurement models and confirmatory factor analysis (CFA) within structural equation modeling (SEM) is more suitable to test the full model (Fabrigar et al., 1999). In this study, the four measurement models are the first-order latent constructs of emotional value, social value, quality value, and price value. Each first-order latent construct is captured by multiple measures. The full model is a second-order latent CPV model. In high-order modeling, one

major complication could occur when multiple exogenous variables (i.e. first-order latent constructs) all are entered simultaneously in a single structural model. The significant effects on a dependent variable (i.e. second-order latent construct) shown by exogenous variables individually might vanish or exhibit an opposite than expected sign (Koufteros et al., 2009). There is a high likelihood that some measures might show cross-loadings with non-intended first-order latent constructs. Second-order latent model should recognize the contribution and retain the idiosyncratic nature of each first-order construct and treat such constructs as facets of the higher-order construct. Path coefficients should relate each first-order construct to its second-order construct and therefore both the substantive and statistical contribution of each first-order construct can be estimated and evaluated. (Gerbing et al., 1994; Koufteros et al., 2009). The specific statistical examinations conducted include the model-to-data fit, unidimensionality, reliability, and construct validity (including both convergent validity and discriminant validity). SPSS 16 was employed for EFA analysis. The full model tested by CFA using LISREL 8.8 involved three levels: measures, first-order latent constructs, and a second-order latent construct.

4. Results and discussion

4.1. Descriptive statistical analysis

The profile of survey respondents is summarized in Table 1. Most respondents are female at 67.8%, while male only accounts for 32.2%. This gender distribution pattern is consistent with previous online consumer surveys. Over 60% of respondents were 45 and over, with over one third (34.2%) being 45–54, while 28.1% were aged 55 and above. The next largest category was 35–44 olds (19.1%). The response, therefore, is skewed toward older consumers, who might have more discretionary income compared to younger consumers. With regard to education level, besides 4.1% non-answer returns, 23.5% respondents indicated they were high school graduates and 40.1% had some college education, while four-year college graduates and advanced degree receivers accounted for 22.8% and 9.5%, respectively. Although

Table 1
Profile of survey respondents.

Characteristic	Percent (%)	Characteristic	Percent (%)
Gender		Annual household income	
Male	32.2	< \$20k	15.6
Female	67.8	\$20k–40k	27.7
		\$41k–75k	31.9
		\$76k–100k	11.6
		> \$100k	12.2
Age		Education level	
17 or younger	0.5	High school graduate	23.5
18–24	6.9	Some college	40.1
25–34	11.2	4-year college graduate	22.8
35–44	19.1	Advanced degree	9.5
45–54	34.2	(Master, Ph.D., M.D. J.D., etc.)	
55 and over	28.1	Others	4.1
Race			
Caucasian White	84.3		
African American	7.2		
Hispanic Latino	3.8		
Asian, Pacific Islander	2.9		
Native American,	0.7		
Aleut, or Eskimo			
Others	1.1		

Note: the total eligible returns are 4949.

household income is a sensitive question, most respondents did answer it. The largest proportion of respondents (31.9%) fell into the \$41k–\$75k income category, while 27.7% indicated their income fell between \$20k and \$40k. Nearly one quarter of respondents earned more than \$75k, of which slightly more than half (12.2%) reported they made more than \$100k a year. By contrast, some 15.6% respondents indicated their household income was less than \$20k. The ethnicities of respondents consist of Caucasian White at 84.3%, African American at 7.2%, Hispanic Latino at 3.8%, Asian, Pacific Islander at 2.9%, Native American, Aleut, or Eskimo at 0.7%, and others at 1.1%. Compared to the demographics of national population, Caucasian White showed a higher percentage in this study while the weights of African American and Hispanic Latino respondents were relatively lower.

Table 2 presents respondents' general shopping behavior for casual sportswear. Mass merchandisers and mid-tier retailers are the most common channels where American consumers purchase casual sportswear, accounting for 43.6% and 23.3%, respectively, followed by specialty stores at 11.8% and department stores at 10.4%. As for purchase frequency, the largest proportion of respondents (34.8%) indicated they bought casual sportswear only on special occasions. Some 33.2% shopped once a month on average, while only 12.3% shopped more frequently. With

regard to average monthly expenditure on casual sportswear, about half of respondents indicated they spent less than \$25 with a further 26.5% spending \$25–50. Some 18.6% spent \$51–\$100, while only 3.7% of respondents spent more than \$100 per month. The latter might represent the consumers of high-end sportswear. When asked about fiber content and attitude toward cotton, 69.1% respondents indicated they did check the fiber content before making a purchase and an overwhelming 84.1% of respondents claimed that in general they liked cotton fabric more than fabrics made from other fibers.

Table 3 shows the attitudes of survey participants toward casual sportswear and the cotton usage in casual sportswear. The responses were structured using a seven-point Likert scale. In general, most respondents enjoyed shopping for casual sportswear. The importance of fashionable style was perceived differently by the respondents as they had different desires. Based on the responses, product/company and brand reputations are critical for casual sportswear to attract and retain customers. The propensity for shopping in the same stores and staying with the same brands could create barriers to new entrants. However, the mentality of best value for money shown by a big portion of respondents could provide opportunities for those companies that gain cost advantage. The responses revealed overwhelmingly positive attitudes toward

Table 2
Respondents' general shopping behavior for casual sportswear.

Question	Response (%)	Question	Response (%)
Where is your most favorite place to shop for casual sportswear		Averagely, how often do you shop for casual sportswear	
Mass Merchants (Wal-Mart, Target, Kmart, etc.)	43.6	More than once a week	1.6
		Once a week	3.1
Mid-Tier Retailers (Kohl's, JC Penney, Sears, etc.)	23.3	Once every other week	7.6
		Once a month	33.2
		Special occasions	34.8
		Others	19.7
Department Stores (Macy's, Dillard's, Belk, etc.)	10.4	How much, averagely, do you spend on purchasing casual sportswear each month	
		< \$25	51.2
Specialty Stores (Sport Authority, Old Navy, Gap, etc.)	11.8	\$25–50	26.5
		\$51–75	12.2
Others	10.9	\$76–100	6.4
		> \$100	3.7
Do you pay attention to the fiber content when you shop for casual sportswear		In general I like cotton fabric over others	
Yes	69.1	Yes	84.1
No	30.9	No	15.9

Note: the total eligible returns are 4949.

Table 3
Attitudes toward casual sportswear and the cotton usage in casual sportswear (in %).

	Extremely unimportant					Extremely important	
	1	2	3	4	5	6	7
I enjoy shopping casual sportswear	4.8	4.9	8.7	25.4	21.1	21.3	13.8
Fashionable style is very important to me	5.1	11.7	13.3	25.7	17.2	10.1	14.3
I usually purchase casual sportswear from reputable apparel brands	3.6	3.3	7.5	27.5	24.5	22.1	11.5
I always go to the same store/stores to shop casual sportswear	5.8	7.5	14.5	27.2	24.6	13.0	7.4
I tend to stick to the apparel brands I like for buying casual sportswear	5.9	5.7	10.9	28.3	25.3	16.1	7.8
The more apparel product information I learn, the harder it seems to choose the best	13.9	14.9	19.4	31.6	10.7	5.5	4.0
I carefully calculate how much I spend on shopping casual sportswear	4.4	5.0	8.7	26.8	22.2	19.2	13.7
I look carefully to find the casual sportswear with the best value for money	2.2	1.9	3.9	20.4	25.8	26.3	19.5
Casual sportswear made with cotton fabric is likely to be more fashionable than others	3.9	4.7	9.1	45.5	17.5	8.1	11.2
Casual sportswear made with cotton fabric is likely to be of higher quality than others	3.2	3.8	7.3	42.3	20.8	14.2	8.4
Casual sportswear made with cotton fabric is likely to be more functional than others	2.7	2.8	5.5	35.4	24.2	18.5	10.9
Casual sportswear made with cotton fabric is likely to be more pleasant to the skin than others	2.4	1.9	3.5	27.6	25.1	23.0	16.5
Casual sportswear made with cotton fabric is likely to be more comfortable than others	2.3	2.0	3.3	26.9	24.6	23.1	17.8

Note: the total eligible returns are 4949.

cotton usage in casual sportswear. For instance, the majority of respondents agreed that cotton fabric is likely to be more fashionable than other fabrics, and casual sportswear made of cotton fabric is likely to be of higher quality, more functional, more pleasant to the skin, and more comfortable compared to other types of fabrics. These results partly explain the rapid growth of cotton consumption in casual sportswear.

4.2. Results of EFA and CFA of CPV model

The non-response bias testing on demographic information showed there were no significant differences between early and late groups of returned surveys. The measures—*has poor workmanship* and *would not last a long time* in the quality value construct and *would make me want to wear it* in the emotional value construct were dropped due to high cross loading and low factor loading respectively.

Table 4 summarizes the final results from EFA. The results suggested that all four first-order latent constructs met the unidimensionality criterion. The measures captured four distinct dimensions and the individual measures contributed to the expected construct. The eigenvalues for each factor all were above cutting-point one. The four constructs cumulatively accounted for some 77% of the variance in CPV of casual sportswear. This is very satisfactory. Price value explained the most variance of CPV at some 40%, followed by quality value, emotional value, and social value at 19.4%, 11.9% and 5.4%, respectively. As Cronbach's coefficient alphas were all above criterion value 0.7, the evidence of reliability was established.

Appendix A presents the results of convergent validity, discriminant validity, and model-to-data fit tests for all first-order latent constructs. All of the Average Variance Extracted (AVE) scores were above the desired threshold of 0.5, which indicates the criterion of convergent validity was met. None of the confidence interval of 2 standard errors around the correlation between each respective pair of constructs captured 1.0, indicating the criterion of discriminant validity was met. The results also show that all constructs met the model-to-data fit requirements. The establishment of unidimensionality, reliability, and validity of

constructs and the model-to-data fit make perceptual measures viable and dependable in large-sample empirical studies (Chen and Paulraj, 2004). The results show that the developed survey instrument could generate reliable data.

Fig. 2 illustrates the second-order CFA models for CPV of casual sportswear, including the standardized factor loadings and corresponding *t*-values. The final CFA model showed an excellent fit to the collected consumer survey data. The four first-order latent constructs designed to measure CPV, namely price value, social value, emotional value, and quality value, all exhibited high and significant factor loadings on CPV. In particular, price value and quality value showed relatively greater influence on CPV of casual sportswear. These results are consistent with the finding from the previous EFA that price value and quality value accounted for more variance of CPV compared to social and emotional values.

5. Conclusions and implications

In the past decade, casual sportswear consumption has been witnessing an extraordinary growth in the U.S. The acceptability

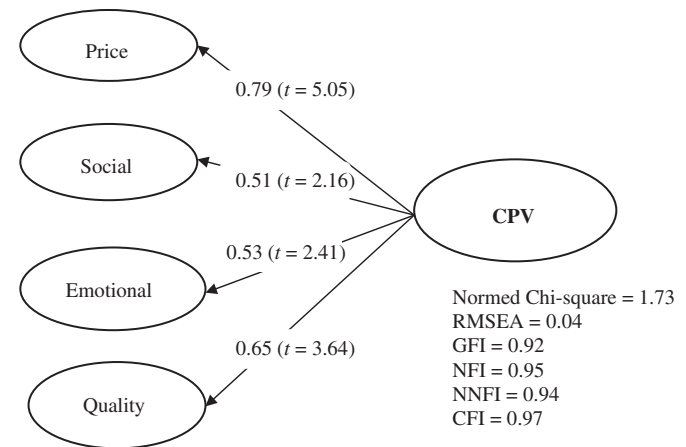


Fig. 2. Results of the second-order CFA models for CPV of casual sportswear.

Table 4
Results of EFA.

	Factor 1	Factor 2	Factor 3	Factor 4
Price				
1. Is reasonably price	0.888	0.055	0.124	−0.062
2. Offers value for money	0.901	0.060	0.152	−0.066
3. Is a good product for the price	0.903	0.079	0.166	−0.058
4. Would be economical	0.822	0.114	0.074	−0.079
Social				
1. Would help me to feel acceptable	0.241	0.853	0.028	0.008
2. Would improve the way I am perceived	0.158	0.923	0.022	−0.004
3. Would make a good impression on other people	0.144	0.922	0.046	−0.006
4. Would give its owner social approval	0.069	0.906	0.011	0.001
Emotional				
1. Is one that I would enjoy	0.216	0.250	0.761	0.140
2. Is one that I would feel relaxed about using	0.205	0.209	0.789	0.115
3. Would make me feel good	0.328	0.160	0.763	0.134
4. Would give me pleasure	0.406	0.115	0.683	0.126
Quality				
1. Has consistent quality	0.109	0.440	−0.006	0.696
2. It is well made	0.080	0.451	0.001	0.737
3. Has an acceptable standard of quality	0.125	0.378	0.009	0.645
4. Would perform consistently	0.123	0.330	0.076	0.691
Cronbach's coefficient alpha	0.957	0.939	0.927	0.905
Percent of variance explained	39.933	5.372	11.860	19.401
Cumulative percent	39.933	45.305	57.165	76.566

of casual dress on more occasions has paved the way for sportswear to move into the mainstream apparel market. Given the size of the business and its promising future, this study represents one of the very first empirical research studies devoted to understanding the perceived value of casual sportswear from the American consumers' perspective. The findings enhance knowledge of the CPV concept and its measurement at a product level. They provide some useful insights for both academic researchers as well as industrial practitioners.

Overall, this study contributes to the literature in four ways. First, based on previous conceptual and empirical research, it proposes a CPV analytical model in four distinct constructs—*price value, quality value, social value, and emotional value* and has developed corresponding measures and scales for casual sportswear. Second, using primary data gathered from a national consumer survey, it statistically assessed the unidimensionality, reliability, validity, and model-to-data fit of the CPV model and proved the model is valid, the measurement is adequate, and the survey instrument can generate reliable data. Moreover, these four constructs all showed significant impacts on CPV and captured most of the variance of CPV regarding casual sportswear. The accountabilities of each construct on the variance of CPV were also quantitatively determined. Third, beyond descriptive analysis and subjective judgment, the results of CPV analysis provided rigorous statistical evidence unraveling the consumers' attitudes and shopping behavior toward casual sportswear. The value based knowledge derived from this study enables more effective understanding of consumer product search and purchase decision-making process. Finally, there is a reciprocal relationship between apparel market and cotton industry. Given the recent fact of high demand on cotton made sportswear, this study addressed the links between the desired values identified by this CPV analysis and the values delivered by cotton made sportswear. Value creation is the key for companies to achieve long-term success.

This study also imparts several research and managerial implications. For academia, this study added knowledge for future studies exploring CPV and its possible relationships with other key components in consumer behavior. The four dimensional CPV model has sound and stable psychometric properties while the parsimony criterion is also well met. The results demonstrate that CPV is not a unidimensional construct and consumers assess products, not just from utilitarian aspects such as price value and quality value, but also in terms of the emotional value (e.g., the enjoyment and pleasure derived from the product) and the social consequences of what the product communicates to others. Although the measures and scales were tested in a context of casual sportswear, the methodology may, therefore, be applied to other types of products or services. In addition, the cooperation with a professional market research company enabled the survey to reach a large consumer database and significantly increase the number of survey responses. A large sample of respondents better reflects the population and makes the findings from this study more valuable and applicable.

For industrial practitioners, recognition of the importance of the different dimensions of consumer desired value could assist cotton producers and apparel manufacturers in developing more competitive products and allow retailers to reach target consumers using more effective marketing and promotional strategies. Although social value consideration and emotions play a part in every purchase decision, price has been revealed as the most influential factor in this study while quality is determined as the secondary important value among the CPV of casual sportswear. The focus of value for money well explains the consumers' shopping behavior and attitudes identified in Tables 3 and 4. Price competitive retailers including mass merchandisers (e.g., Wal-Mart, Target, etc) and mid-tier retailers (e.g., Kohl's, JC Penney, etc.) are the most popular places for the majority of consumers to purchase casual sportswear. For most

consumers, given the limited budget, they want more information regarding a product so they can make better decision for their money. Fiber content is one of the key pieces of information, which consumers look for while shopping. The usage of cotton fabric or cotton blended fabric adds advantages to casual sportswear due to cotton's perceived features of higher quality, more functional, more comfortable, better for body, skin, etc. Today, it becomes critical for the U.S. cotton industry to work closely with textile and apparel manufacturers to develop quality casual sportswear at more affordable prices. In the last few years, due to the worsening economic conditions and record-high unemployment rate, many U.S. retailers have experienced a sharp downturn in sales and eroding profit margins. The shift of promotional and retail sales orientations back to an emphasis of value for money and overall quality rather than hedonic dimensions of value seems particularly useful and more appealing to the majority of consumers. In addition to offering a competitive price points, recognized brands are exclusive assets for the U.S. casual sportswear manufacturers and retailer in their battle for better customer loyalty and higher profits, since consumers usually consider respected brands mean dependable quality. A significant body of consumers tends to purchase casual sportswear from some brands and certain stores. This shopping habit protects the extant branded apparel companies and retailers but sets up a challenging barrier for the potential non-known new entrants.

Overall, it would be expected that if a product could deliver a consumer's desired value, he or she would be more willing to buy the product, be more willing to recommend the product, and show greater satisfaction with the product.

6. Limitations and future studies

Although this study adds rigorous empirical evidence to the CPV literature and provides a series of implications to both academicians and industrial practitioners, there are still several limitations that need to be mentioned and also can be considered as possible directions for future studies.

First of all, perceived product value should be viewed as a part of a continuous process in the maintenance of a relationship between apparel manufacturer and retailer with target customers. This research provides a measure of what U.S. consumers perceived important for casual sportswear at a particular time point. Future studies employing a longitudinal view may shed different light on the relative importance and perhaps the susceptibility of change of the four dimensions of value. Second, as an exploratory study, this research was only focused on CPV of casual sportswear. In the future, the relationships between CPV and other constructs such as consumer satisfaction and purchase intentions can be examined. Finally, although four dimensions—price value, quality value, social value, and emotional value can capture most variance of CPV for casual sportswear, some underlying factors (e.g., epistemic value, conditional value etc.) that contribute to the unexplained variance in the model can be further explored.

Appendix A. The auxiliary results

See appendix Tables A1 and A2.

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Table A1

Results of convergent validity and discriminant validity tests.

Construct	AVE score	Construct	Correlation	Std. error	Confidence interval
Price	0.725	Price–Social	0.177	0.110	(–0.043, 0.397)
Social	0.580	Price–Emotional	0.140	0.100	(–0.060, 0.340)
Emotional	0.597	Price–Quality	0.351	0.110	(0.131, 0.571)
Quality	0.694	Social–Emotional	0.022	0.110	(–0.198, 0.232)
		Social–Quality	0.032	0.120	(–0.208, 0.272)
		Emotional–Quality	0.333	0.120	(0.093, 0.573)

Table A2

Results of goodness-of-fit indices.

Construct	Normed χ^2	RMSEA	GFI	NFI	NNFI	CFI
Price	1.53	0.03	0.95	0.99	0.98	0.99
Social	1.85	0.07	0.91	0.95	0.94	0.95
Emotional	1.76	0.06	0.92	0.94	0.95	0.98
Quality	1.25	0.04	0.94	0.97	0.97	0.99
Recommended values	≤ 0.2	≤ 0.08	≥ 0.90	≥ 0.90	≥ 0.90	≥ 0.90

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